

InterviAI Interview Report

AI-powered mock interview assessment

CANDIDATE

hasan arsalan

hasanarsalan781@gmail.com

ROLE

**Backend
Developer**

DIFFICULTY

Easy

DATE

August 16, 2026

RECOMMENDATION

Hire

Scores

7.4 OVERALL	7.5 TECHNICAL	7.0 COMMUNICATION	6.5 CONFIDENCE	8.5 PROBLEM SOLVING
-----------------------	-------------------------	-----------------------------	--------------------------	-------------------------------

Strengths

- Solid grasp of fundamental relational database concepts including primary/foreign keys, joins, and indexing trade-offs.
- Strong problem-solving capability, providing an optimal $O(n)$ time complexity Hash Map solution for Two Sum.
- Clear understanding of core data structure trade-offs between Hash Maps and Binary Search Trees.
- Articulate explanation of personal motivation for backend engineering and a clear roadmap for technical growth.

Weaknesses

- Failed to describe project implementation details and database schemas when explicitly prompted about resume projects.
- Gave filler/evasive responses during initial project deep-dive questions before technical pivoting.
- Behavioral debugging scenarios lacked specific technical details regarding FastAPI and database connection logic.

Highlights

- Clear explanation of database indexing benefits and trade-offs regarding INSERT/UPDATE performance.
- Flawless, immediate response and complexity analysis for the Two Sum problem.

Concerns

- Avoidance of direct questions about project architecture raises minor flags about candidate depth or familiarity with claimed resume projects.

Technical Evidence

- Explained Primary Key vs Foreign Key differences using concrete student/enrollment table examples.
- Differentiated INNER JOIN and LEFT JOIN including NULL handling for non-matching records.
- Explained database B-tree index query speed improvements alongside write performance and storage overhead.
- Compared Hash Map average $O(1)$ complexity against BST average $O(\log n)$ and worst-case $O(n)$ operations.
- Proposed an optimal $O(n)$ time and $O(n)$ space complexity Two Sum algorithm using a Hash Map.

Learning Roadmap

- Practice presenting personal project architectures, explicitly explaining ER diagrams, table schemas, and technical trade-offs.
- Deepen knowledge of FastAPI backend concepts, such as asynchronous database drivers (SQLAlchemy/Alembic) and dependency injection.
- Gain hands-on experience implementing enterprise-level backend patterns, including JWT/OAuth2 authentication and API rate limiting.
- Study real-world database optimization techniques beyond basic indexing, such as query explain plans, connection pooling, and caching with Redis.

Competency Report

Introduction

Strong

Clear overview of technical stack and genuine interest in backend development logic.

Projects

Weak

Unable to provide specific database schema or structural details for the claimed flight delay project when prompted twice.

Technical Knowledge

Strong

Demonstrated solid core knowledge of SQL joins, keys, indexing overhead, and data structure lookup trade-offs.

Problem Solving

Strong

Accurately identified optimal $O(n)$ time and space complexity solution for Two Sum using a Hash Map.

Behavioral / Debugging

Average

Described a standard methodology for resolving environment configuration bugs in FastAPI, though details remained surface-level.

Teamwork

Average

Explained a reasonable conflict-resolution approach for API design and basic task delegation in academic group projects.

Motivation & Career Goals

Strong

Articulated a clear path for growth in system design, FastAPI, databases, and scalable backend infrastructure.

Overall Feedback

Hasan demonstrated strong foundational computer science knowledge suitable for an entry-level Backend Developer. While he stumbled early on by failing to articulate the database schema for his personal flight delay project, he recovered effectively when assessed on standard SQL, indexing, and data structure concepts. His problem-solving was strong, delivering the optimal solution to the Two Sum problem with correct complexity analysis. Communication was generally clear, and his motivation for backend development aligns well with junior engineer expectations. With appropriate guidance on real-world production architectures, Hasan is well-prepared to contribute to a backend team.